# Lensing of moiré patterns by atomic area defects: topological node charge, dressed charge, and the ring-traction identity

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A point defect in one layer of a bilayer—a vacancy cluster, an inserted island, or a locally rotated patch—bends the moiré domain-wall network over distances far larger than itself and can remove moiré nodes. We ask, within a continuum model of the relaxed registry, how many nodes a defect removes and where they go. AA nodes are the degree density of the registry map $u : \Sigma \to T^2$ and obey a Gauss law in which every defect core carries an integer charge. An area or torque defect with complex charge $C$ deforms the moiré through the lens map $\beta = z - m/\bar{z}$ with mass $m = C/\kappa$, where $\kappa = \theta e^{i\alpha}$ encodes twist or mismatch; for real $m$ this is the point-mass gravitational lens. Averaged over the position of the defect in the moiré cell, the removed node charge $\theta n_{\rm miss}$ splits into a far share $\propto \cos\varphi_m$ and a core-bound share $\propto 1 - \cos\varphi_m$, with $\varphi_m = \arg C - \alpha$. Relaxed simulations bind the expected integers for seven of eight defect types and show that the far charge is set by a dressed charge $C_{\rm eff} = \sum C + C_{\rm reg}$, where $C_{\rm reg}$ is the force dipole of the reorganized registry forces and depends on the integer branch. Although the moiré crystal has a negative bulk modulus (twist) or rotational stiffness (mismatch), an exact ring-traction identity shows that local sources are driven by the atomic traction on their boundary, not by these constants. For Einstein radii beyond the moiré period the lens resolves into arcs and, for $\varphi_m = 0$, a ring of uniform stacking; relaxation turns these into spiral or concentric wall textures.

## I. INTRODUCTION

Two lattices stacked with a small twist angle or a small lattice mismatch form a moiré superlattice whose period exceeds the atomic spacing by a factor $1/\theta$. Since the discovery of correlated and superconducting phases in magic-angle graphene [1–3], moiré systems have become a platform for engineering band structures and interactions [4, 5]. The relevant length scale is set by the *relaxed* moiré pattern: interlayer adhesion turns the smooth interference pattern into AB/BA domains separated by shear solitons that meet at AA nodes [6–13]. Because the moiré magnifies the relative displacement of the two layers by $1/\theta$, it is extremely sensitive to strain. Twist-angle disorder, heterostrain, and local strain fields are visible directly in scanning-probe and diffraction maps [14–19].

Point defects are an equally natural source of local strain. Vacancy clusters, ion-irradiation craters, adatom patches and grain-boundary fragments are common in two-dimensional crystals [20]. A cluster of a few tens of missing atoms produces a long-ranged displacement field of order $|C|/r$. In the moiré pattern this field is amplified by $1/\theta$ into a distortion comparable to the moiré period. Two questions then arise: *how many moiré nodes does an atomic defect remove, and where does that missing node charge reside?* The answer is not obvious. The number of missing atoms is an integer, the number of moiré nodes is an integer, and yet the conversion factor between them is the continuous parameter $\theta$.

Related questions have been studied for extended defects and smooth strain. Atomic dislocations in bilayer graphene appear as dislocations of the AB/BA domain pattern [7, 31]. Long-wavelength deformations of the moiré have been described by moiré phonons and phasons [32, 33], and uniform heterostrain reshapes the pattern and the bands [15, 16, 34]. Point defects with a monopole area or torque charge have, to our knowledge, not been treated from the viewpoint of the topology of the registry map.

In this paper we answer these questions within a minimal continuum model. Four results stand out.

(i) *Moiré nodes are topological charges.* The moiré pattern is a map $u$ from the sample to the registry torus $T^2$. AA nodes are the points where $u$ takes the AA value, counted with the sign of $\det \nabla u$; they are the degree density of $u$. A Gauss law fixes the node count in any region from boundary data alone, and every hole carries an integer charge.

(ii) *An atomic area defect acts as a lens for the moiré.* The far field $-C/\bar{z}$ of a defect with complex charge $C$ (area deficit plus torque) maps the undeformed moiré coordinate $\beta$ to the physical coordinate $z$ through $\beta = z - m/\bar{z}$ with lens mass $m = C/\kappa$. For real $m$ this is the point-mass gravitational lens equation [21, 22]; a complex $m$ adds a rotational deflection without gravitational counterpart. Undeformed moiré nodes play the role of light sources. Sources inside a disk of radius $b_s = 2R|\sin(\varphi_m/2)|$ have no images and are swallowed by the defect. As a result, the removed node charge $2\pi\theta|C|/A_c$ splits, on average over the defect position, into a far share $\propto \cos\varphi_m$ and a local share $\propto 1 - \cos\varphi_m$, both fixed by the single phase $\varphi_m = \arg C - \alpha$.

(iii) *The far charge is dressed by the moiré.* The registry forces redistribute when the defect reorganizes the wall network, and their force dipole adds an induced charge $C_{\rm reg}$. The far node charge is fixed by

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$C_{\rm eff} = \sum C + C_{\rm reg}$, not by the atomic charge alone. $C_{\rm reg}$ depends on the integer branch selected by the pattern, and in the one pair we tested at two stiffnesses it behaves more like a displacement-controlled than a force-controlled source.

(iv) *Local sources do not couple through the moiré elastic constants.* We measure the elastic constants of the moiré crystal. The twisted moiré has a negative bulk modulus $K_M < 0$, the mismatch moiré a negative rotational stiffness $\chi_M < 0$. One might expect the response to a local moiré-scale source to follow from these constants. An exact ring-traction identity shows instead that such a source is driven only by the *atomic* traction on its boundary, not by the moiré pre-stress. The far node charge is then accounted for exactly by integer bookkeeping and the force-dipole dressing, although $C_{\rm reg}$ itself is not yet predictable.

Figure 1 shows which atomic defects act as lenses, and Fig. 2 summarizes the central picture. Defects that change the transmitted area or rotation of the layer—vacancy clusters, inserted islands, rotated patches—carry a monopole charge $C$ and lens the moiré. Area-neutral combinations produce only short-ranged distortions, and dislocations are a topologically different object: they create a moiré dislocation rather than a lens.

Section II introduces the model, Sec. III the topology of the registry map, and Sec. IV the lens map and the partition theorem. Section V treats several defects and the conservation of far charge. Section VI introduces the dressed charge. Section VII discusses moiré elasticity and the ring-traction identity. Section VIII treats lenses larger than the moiré period, and Sec. IX discusses experimental consequences.

## II. MODEL

The dynamical variable is the interlayer relative displacement $u(\boldsymbol{r}) \in \mathbb{R}^2$; we use complex notation $z = x+iy$, $u = u_x + iu_y$. The energy is

$$E[u] = \int_\Sigma \Big[\mu\, \varepsilon_{ij}\varepsilon_{ij} + \frac{\lambda}{2}\varepsilon_{kk}^2 + V(u)\Big] \mathrm{d}^2 r, \qquad (1)$$

with $\varepsilon = \frac{1}{2}(\nabla u + \nabla u^T)$ the strain of the total relative displacement. The background $u_b = c_{\rm AA} + \kappa z$ with $\kappa = \theta e^{i\alpha}$ describes a twist ($\alpha = 90°$) or an isotropic lattice mismatch ($\alpha = 0$); we write $\delta u = u - u_b$. In this reference the twisted background (a rotation) is stress-free, while the mismatch background carries a uniform dilational prestress. Appendix A shows when the choice of reference matters. We take $u = u_{\rm top} - u_{\rm bottom}$ and place all defects in the top layer. A defect in the bottom layer enters with $C \to -C$, i.e., $\varphi_m \to \varphi_m + 180°$. For the mismatch, $\kappa = \theta > 0$ means that the top layer has the larger lattice constant.

The registry potential is

$$V(u) = V_0 \sum_{\boldsymbol{G}} \big(1 - \cos \boldsymbol{G}\cdot u\big) + V_3 \sum_{\boldsymbol{G}} \sin \boldsymbol{G}\cdot u, \qquad (2)$$

where $\boldsymbol{G}$ runs over three reciprocal vectors of the triangular lattice $\Lambda$ (lattice constant $a = 1$, cell area $A_c = \sqrt{3}/2$) and $V_3 = \sqrt{3}V_0$. This honeycomb-type registry has a maximum at $c_{\rm AA} = (a_1 + a_2)/3$ (AA stacking) and degenerate minima at AB and BA. It is $C_6$ symmetric about AA, including layer exchange $u \to -u$, and has a mirror symmetry [6, 8].

*Iris defects.* A defect at $z_k$ is a disk of radius $r_h$ removed from the energy. A penalty fixes the mean normal and tangential displacement of its rim, $\langle \delta u \cdot n\rangle = -\mathrm{Re}\, C/r_h$ and $\langle \delta u \cdot t\rangle = -\mathrm{Im}\, C/r_h$, while the rim shape is free. The complex charge $C$ combines an area deficit ($C > 0$, a "closed iris"), an inserted area ($C < 0$), and a rim rotation or point torque ($\mathrm{Im}\, C \neq 0$). In an infinite medium the equilibrium field is

$$\delta u = -\frac{C}{\bar z - \bar z_k}. \qquad (3)$$

It is the gradient of a harmonic potential, so $\mathrm{div}\, \delta u = \mathrm{curl}\, \delta u = 0$ and it solves Eq. (1) for any $\lambda, \mu$ (pure shear). A defect that removes $n_{\rm miss}$ unit cells and is fully closed by the surrounding layer has $C = n_{\rm miss} A_c/2\pi$, as measured by the displacement flux far away. With $|C| = 4$ and $r_h = 2.4$ the core strain is of order one, so the integers bound to a core depend on the nonlinear core and are model specific.

*Numerics.* We minimize Eq. (1) with L-BFGS [29] on a square grid of spacing $h = 0.45$ with $u = u_b$ on the outer boundary. The standard box has half-width 72; the large boxes have half-width 148.5 (grid $661^2$, about 105 nodes). Unless noted, $\theta = 1/30$, $V_0 = 0.005$, $\mu = 1$, $\lambda = -0.5$ (2D bulk modulus $K = \lambda + \mu$, $K/\mu = 0.5$), $|C| = 4$, $r_h = 2.4$. The soft bulk modulus is a deliberate choice that separates the shear and dilation sectors; the dressed-charge calculations of Sec. VI were repeated at $K/\mu = 2$. The registry strength gives domain walls with a half-maximum width of about 4 (in units of $a$) in $V$, i.e., $\approx 0.13 L_M$: the moiré is strongly relaxed. The lens (Einstein) radius is $R = \sqrt{|C|/\theta} \approx 11$ and the relaxed moiré period is $\approx 31$. All parameters are collected in Table VI (Appendix D).

## III. TOPOLOGY OF THE REGISTRY MAP

### A. Degree and the wall network

An affine registry $u = c + Az$ ($\det A \neq 0$) maps the moiré lattice $\Lambda_M = A^{-1}\Lambda$ onto $\Lambda$. It therefore induces a homeomorphism $\mathbb{R}^2/\Lambda_M \to T^2$ of degree $\mathrm{sign}\det A$; in complex notation $u = \kappa z + \eta \bar z$ gives $\det A = |\kappa|^2 - |\eta|^2$. Twist and isotropic mismatch have degree $+1$; a strain dominated by $\bar z$ has degree $-1$.

Cutting $T^2$ into AB and BA regions gives a minimal cell decomposition (a $\Delta$-complex): one vertex (AA), three edges, and two faces, with Euler number zero [Fig. 2(a)]. In the universal cover $\mathbb{R}^2$ the one-skeleton is the arrangement $\mathcal{L}$ of three families of lines through the

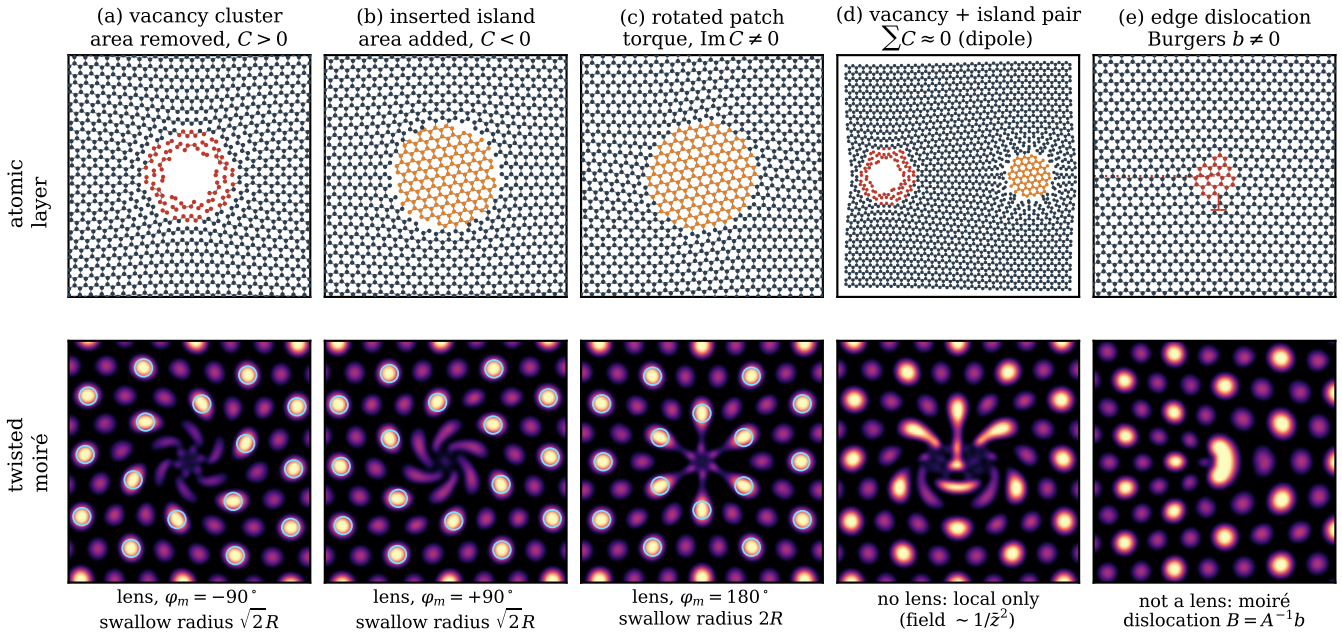



FIG. 1. Which atomic defects lens a moiré pattern. Top: one graphene layer containing (a) a partially closed vacancy cluster ($C > 0$), (b) an inserted island with extra atoms ($C < 0$), (c) a rotated patch ($\mathrm{Im}\, C \neq 0$), (d) a vacancy cluster and an island with $\sum C \approx 0$, and (e) an edge dislocation (Burgers vector $b$; dotted line: cut). Atoms are displaced by the continuum field with far tail $-C/\bar{z}$; highlighted atoms belong to the defect. Bottom: moiré formed with a second, perfect layer twisted by $\theta = 1/30$, computed directly from the atomic positions (bright: atoms on top of atoms, i.e., AA stacking; no interlayer relaxation). Circles mark the node positions predicted by the lens map, Eq. (8). Defects (a)–(c) are lenses with Einstein radii $R \approx 11$–$14$; (d) distorts the pattern only locally; (e) inserts a moiré dislocation $B = A^{-1}b$.

AA lattice points. We define the domain walls as $u^{-1}(\mathcal{L})$ and the nodes as $u^{-1}(\mathrm{AA}{+}\Lambda)$. Two consequences follow:

*Even valence.* The lift of a closed loop with zero Burgers class is a closed curve in $\mathbb{R}^2$, so by the Jordan curve theorem it crosses each line of $\mathcal{L}$ an even number of times. Hence the number $z$ of walls meeting at a vertex, or crossed by a loop around a defect core, is even. Ordinary AA nodes have $z = 6$; the eight- and ten-wall vertices seen near defects obey the same rule. Where all faces are triangles, Euler's theorem gives $\sum_v (6 - z_v) = 6\chi$, so a core behaves like a vertex carrying disclination charge $6 - z_{\mathrm{core}}$ [23, 24].

### B. Node charge and Gauss law

A loop $\gamma$ with vanishing Burgers class $\oint_\gamma \mathrm{d}u \in \Lambda$ lifts to $\mathbb{R}^2$; iris defects have zero Burgers class. We define the node charge

$$Q(\gamma) = \sum_{n\in\Lambda} \mathrm{wind}\big(\tilde{u}\circ\gamma,\ c_{\mathrm{AA}} + n\big) \in \mathbb{Z}, \tag{4}$$

which is invariant under deformations that do not cross a node. At a regular point the index of a node is $\mathrm{sign}\det\nabla u = \pm 1$: ordinary nodes $N_+$ and inverted nodes $N_-$. For a region $D$ containing holes $H_k$ (defect cores) on whose complement $u$ is continuous,

$$N_+(D) - N_-(D) + \sum_k Q(\partial H_k) = Q(\partial D), \tag{5}$$

with node density $\rho_N = \det\nabla u / A_c$. The proof uses additivity of winding numbers: interior edges of a fine subdivision cancel, and a cell containing one node contributes its local degree. Because $\rho_N$ is a null Lagrangian, $\partial_t \rho_N + \nabla\cdot j = 0$ holds identically in any continuous dynamics. Nodes therefore enter or leave only through holes, or are created and annihilated in $\pm$ pairs [25]. This is the sense in which moiré nodes are conserved charges.

*Valence–winding bound.* If the AA points enclosed by the lift of $\gamma$ have windings 0 or $\pm 1$ of one sign with total $w$, then

$$z(\gamma) \geq 2L_{\min}(|w|), \qquad z(\gamma) \in 2\mathbb{Z}, \tag{6}$$

where $L_{\min}(n)$ is the minimum number of lattice lines of the three families needed to cover $n$ triangular-lattice points ($L_{\min} = 3, 5, 6, 7, 8, 9, 9$ for $n = 1, \ldots, 7$). By a Jordan-curve argument, each line through an enclosed point is crossed at least twice. The bound is $2L_{\min}$ rather than $6|w|$ because neighboring points share lines.

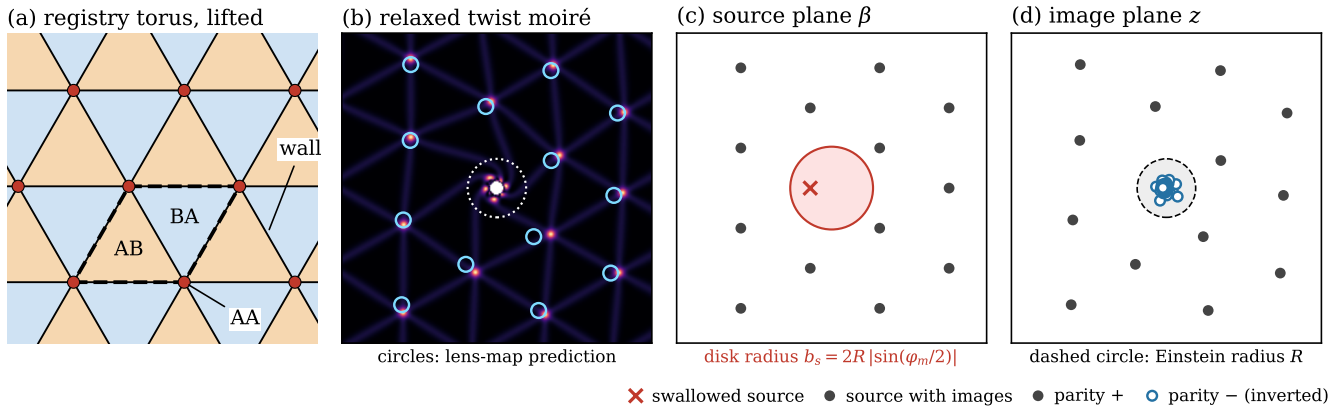



FIG. 2. Moiré nodes as lensed images. (a) Universal cover of the registry torus $T^2 = \mathbb{R}^2/\Lambda$. The three line families $\mathcal{L}$ are the domain walls, the up (down) triangles are AB (BA) stacking, and the lattice points are AA. The dashed parallelogram is one fundamental cell. (b) Relaxed twisted moiré ($\theta = 1/30$) around an area-removing defect ($C = 4$, white disk); color shows the registry energy $V(u)$. Circles mark the positive-parity images predicted by the lens map; the dotted circle is the Einstein radius $R = \sqrt{|C|/\theta}$. (c) Source plane $\beta$: undeformed node positions. Sources inside the disk of radius $b_s$ (crosses) have no images. (d) Image plane $z$ for the same lens ($\varphi_m = -90^\circ$). Each surviving source has an ordinary image outside $R$ (filled) and an inverted image inside $R$ (open).

*Counting.* The discrete count used throughout (Appendix C) satisfies Eq. (5) exactly on the grid, so the zero deviation over 84 relaxed states [Fig. 3(a)] is a consistency check of the counting rather than an independent test. A single area-removing defect in the twisted moiré has $Q_{\text{hole}} = 5$ and $N_- = 4$, so it binds $q \equiv Q_{\text{hole}} - N_- = +1$ node; in the mismatch moiré $Q_{\text{hole}} = N_- = 2$, so $q = 0$ [Fig. 3(b)]. Among 444 circles around cores, 162 saturate the bound of Eq. (6) [Fig. 3(c)]: walls around cores are close to minimal.

## IV. AREA DEFECTS AS MOIRÉ LENSES

### A. Complex charge

For any loop around a set of defects, Eq. (3) gives

$$\oint_\gamma \delta u \cdot (n + i\,t)\,\mathrm{d}s = -2\pi \sum_k C_k. \tag{7}$$

Thus $C$ is a conserved charge, and $C$ and $-C$ can be created or annihilated in pairs. However, the field (3) is traceless and curl-free, so it does no first-order work on a dilatation or torque center. Two such charges therefore have no long-range elastic interaction regardless of their signs, as for Eshelby inclusions in an infinite medium [26]. Elasticity alone is sign-blind.

### B. Lens map

Writing $u = c + \kappa\beta$ defines the undeformed moiré coordinate $\beta$. For one defect at the origin,

$$\beta(z) = z - \frac{m}{\bar{z}}, \qquad m = \frac{C}{\kappa} = R^2 e^{i\varphi_m}, \tag{8}$$

with Einstein radius $R = \sqrt{|C|/\theta}$ and mass phase $\varphi_m = \arg C - \alpha$. For real $m$ this is the lens equation of a point mass, with source position $\beta$ and image position $z$ [21, 22]; $\varphi_m = 180^\circ$ corresponds to a negative mass. For complex $m$ the deflection $m/\bar{z}$ also rotates the images, which has no gravitational counterpart. We nevertheless keep the lens language, since the image structure is the same. The lens mass $m$ is the defect charge transferred to the moiré level, and the undeformed moiré nodes form a perfect lattice of density $n_M = \theta^2/A_c$ in the source plane. The Jacobian $J = 1 - R^4/|z|^4$ is negative inside the Einstein circle, where images are inverted.

*Image structure.* Let $z = re^{i\phi}$. Then $|\beta|^2 = r^2 + R^4/r^2 - 2R^2\cos\varphi_m$. A source at $|\beta| = b$ therefore has no image if

$$b < b_s = 2R\,\big|\sin(\varphi_m/2)\big|, \tag{9}$$

and otherwise has one ordinary image at $r_+ > R$ and one inverted image at $r_- = R^2/r_+$. The disk $b < b_s$ is a shadow of the lens; it exists only for $\varphi_m \neq 0$. Its size depends on the relative phase of the defect charge and the moiré background. For an area defect in a twisted moiré, $\varphi_m = \mp 90^\circ$ and $b_s = \sqrt{2}R$. For an area-removing defect in the larger-lattice layer of a mismatch moiré, $\varphi_m = 0$ and nothing is swallowed; in the smaller-lattice layer, $\varphi_m = 180^\circ$ and $b_s = 2R$. Figure 2(b) shows that

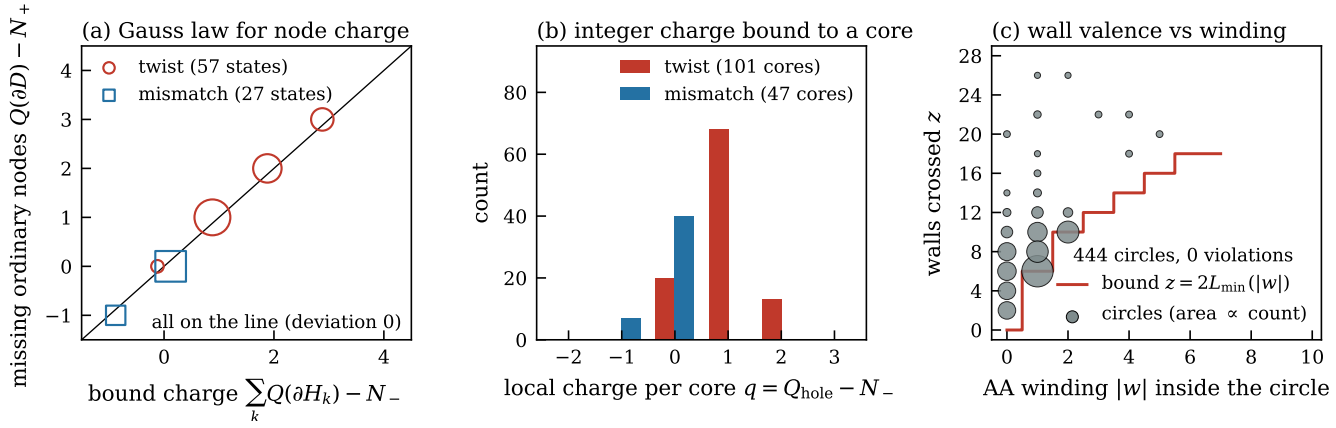



FIG. 3. Topological charges of the registry map. (a) Gauss law, Eq. (5): the ordinary-node deficit equals the charge bound to holes and inverted nodes in all 84 states. Symbol area is proportional to the number of states. (b) Distribution of the integer charge $q = Q_{\rm hole} - N_-$ bound to a single core, over all single-defect and pair states ($K/\mu = 0.5$). (c) Number of walls $z$ (not to be confused with the coordinate $z$) crossed by circles of radius 5, 7, 9 around cores versus the AA winding $|w|$ they enclose. The red staircase is the bound of Eq. (6).

the predicted ordinary images coincide with the relaxed AA nodes outside the core region.

### C. Partition theorem

Averaging over random translations of the moiré lattice, the expected number of sources in a region $S$ of the source plane is $n_M|S|$. Positive images inside $|z| < \rho$ correspond one-to-one to sources with $b_s < b < b(\rho) = |\rho - m/\rho|$, and the loop $|z| = \rho$ maps onto a curve enclosing area $\pi b(\rho)^2$. For $\rho > R$ this yields

$$\Delta N_+(\rho) = T - \frac{\pi n_M R^4}{\rho^2}, \qquad T \equiv \frac{2\pi\theta|C|}{A_c},$$
$$\Delta Q(\rho) = -\frac{2\pi}{A_c}\mathrm{Re}(\bar\kappa C) + \frac{\pi n_M R^4}{\rho^2},$$
$$q = Q_{\rm hole} - N_- = T\,(1 - \cos\varphi_m), \qquad (10)$$

which are tied together by the Gauss law $q = \Delta N_+ + \Delta Q$; here we used $n_M R^2 \cos\varphi_m = \mathrm{Re}(\bar\kappa C)/A_c$. The ordinary-node deficit at infinity, $T = \theta\, n_{\rm miss}$, is independent of $\varphi_m$: *a defect missing $n_{\rm miss}$ unit cells removes $\theta n_{\rm miss}$ moiré nodes.* Of these, the share $T(1 - \cos\varphi_m)$ is swallowed and bound to the core, and the share $T\cos\varphi_m$ is spread to infinity as boundary charge $\Delta Q$. For our parameters $T = 0.967$.

In the relaxed moiré the local share must be an integer, and Eq. (10) becomes a rounding rule whose deviations are carried by the far share. Figure 4 and Table I compare the eight defects $C = \pm 4, \pm 4i$ in both backgrounds. In seven of the eight, the bound integer $q$ equals the rounded prediction. The exception, a torque along the twist sense ($\varphi_m = 180°$), binds $q = 1$ instead of 2; we did not find a $q = 2$ state. Because $T = 0.97$ is close to one, this test mainly confirms the $\varphi_m$ dependence, which also follows from the symmetries below. Testing the magnitude requires a scan in $|C|$; the large lenses of Sec. VIII show that relaxation can shift the bound integer away from the rounded value. The far share $-\Delta Q$ follows $T\cos\varphi_m$ [open symbols in Fig. 4(b)]; the largest deviation is the twisted torque at $\varphi_m = 0$, with $-\Delta Q \approx 0.5$ instead of 0.97.

### D. Symmetry

Layer exchange ($u \to -u$, an inversion about AA) maps $(\kappa, C) \to (-\kappa, -C)$ and preserves $\varphi_m$. The mirror maps $(\kappa, C) \to (\bar\kappa, \bar C)$ and sends $\varphi_m \to -\varphi_m$. Hence (a) in the twisted moiré, removing and adding area ($\varphi_m = \mp 90°$) give identical energies and integer data; (b) in the mismatch moiré, removing and adding area are *not* related by symmetry—they bind 0 and 2 nodes; (c) the two torques in the mismatch moiré are mirror images. All three statements hold in Table I. In particular,

TABLE I. Eight complex defects ($|C| = 4$) in the top layer. "Along twist" means a rotation of the top layer in the sense of the background twist. Predicted local share $T(1-\cos\varphi_m)$, measured bound integer $q$, and defect energy relative to the background.

| defect | $\varphi_m$ | pred. | $q$ | $E - E_{\rm bg}$ |
|---|---|---|---|---|
| mismatch, area removed | 0° | 0.00 | 0 | 14.34 |
| mismatch, torque ± | ±90° | 0.97 | 1, 1 | 14.75, 14.75 |
| mismatch, area added | 180° | 1.93 | 2 | 16.20 |
| twist, torque against twist | 0° | 0.00 | 0 | 14.69 |
| twist, area removed/added | ∓90° | 0.97 | 1, 1 | 15.74, 15.74 |
| twist, torque along twist | 180° | 1.93 | 1 | 18.25 |

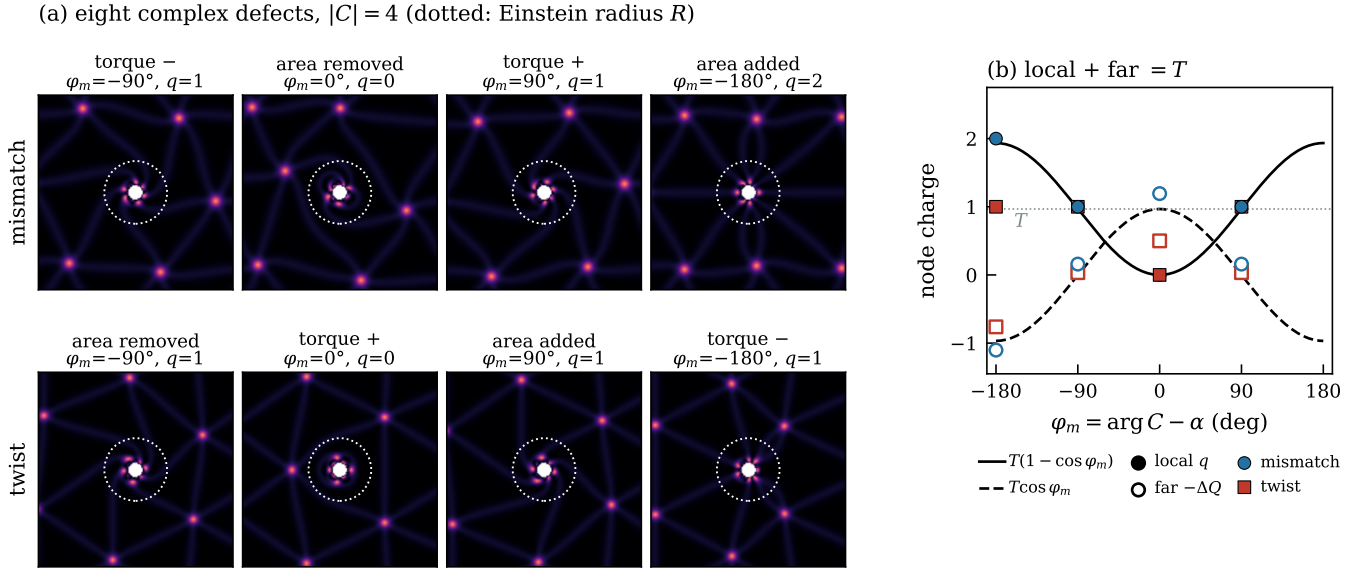



FIG. 4. Partition of the removed node charge. (a) Registry energy around the eight defects $C = \pm 4, \pm 4i$ in the mismatch (top) and twisted (bottom) moiré, ordered by $\varphi_m$. Each title gives $\varphi_m$ and the integer $q$ bound to the core. (b) Local share $q$ (filled) and far share $-\Delta Q$ (open, averaged over $36 < \rho < 58$) versus $\varphi_m$, compared with $T(1 - \cos\varphi_m)$ and $T\cos\varphi_m$, where $T = 2\pi\theta|C|/A_c = 0.967$. Blue circles: mismatch; red squares: twist.

a closing iris and an opening iris carry opposite conserved charge and opposite far charge, yet the same sign of mean bound charge, since $1 - \cos\varphi_m \geq 0$; individual cores in pairs can bind $q = -1$ (Fig. 3(b)).

## V. SEVERAL DEFECTS: CONSERVATION AND ITS LIMITS

For a large loop $\gamma$ enclosing all defects, write $u = u_b + \delta u$ on $\gamma$. To first order in $\delta u$, the area enclosed by the image curve changes by $\theta \oint [\cos\alpha\, \delta u \cdot n + \sin\alpha\, \delta u \cdot t]\mathrm{d}s$. Hence

$$Q(\gamma) - Q_{\mathrm{bg}}(\gamma) = -\frac{2\pi}{A_c}\mathrm{Re}\big(\bar{\kappa}\, C_{\mathrm{eff}}(\gamma)\big), \qquad (11)$$
$$C_{\mathrm{eff}}(\gamma) = -\frac{1}{2\pi}\oint_\gamma \delta u \cdot (n + i\,t)\,\mathrm{d}s,$$

up to terms of order $|m|^2/\rho^2$. Multipoles $e^{-in\phi}$ with $n \geq 2$ do not change the area. If $\gamma$ lies in the linear region and the moiré induced no forces, $C_{\mathrm{eff}} = \sum_k C_k$, and the far node charge would be linear in, additive in, and conserved with the atomic charge.

Figure 5(a,b) tests this on single defects and pairs in the standard box. We compare $\Delta Q(\rho)$ averaged over $36 < \rho < 58$ with the linear prediction from $\sum C$, including the image fields of the clamped box. Over 17 states the slope is 0.81 and the correlation 0.91. The integer $\Delta Q(\rho)$ jumps by $\pm 2$ to $\pm 6$ whenever a symmetric shell of nodes crosses the loop, so individual values carry an uncertainty of about $\pm 0.5$.

*Pairs.* With $V_0 = 0$ (pure elasticity), $(+,-)$ and $(+,+)$ pairs have almost the same interaction energy: $-2.99$ and $-3.14$ at $d = 8$, $+0.06$ and $-0.07$ at $d = 40$ [Fig. 5(c)]. The moiré distinguishes signs only at contact. In the mismatch moiré at $d = 0.73R$ the moiré-mediated energies are $U_{+-} = -1.13$ and $U_{++} = +1.03$, and the $(+,-)$ pair binds no node. Within the lens map the removed node charge is $n_M \int \min(1, |f'|^2)\mathrm{d}^2 z$ with $f' = -\sum_k m_k/(\bar{z} - \bar{z}_k)^2$ (Appendix B). It is subadditive for opposite charges because $|C_1 + C_2| < |C_1| + |C_2|$: the attraction between a closing and an opening iris is a contact interaction of range $\sim 2R$.

In the twisted moiré, $(+,+)$ pairs show several integer branches at the same separation, binding 1, 2, or 3 nodes in total [Fig. 5(d)]. Pulling the cores apart from short distance keeps a "charge-starved" branch (cores bind $(+1, 0)$). In it, a doubled wall tethers the two cores and traps an AA node that neither core absorbs [Fig. 5(e)]. Pushing them together from large distance keeps the branch in which each core binds one node [Fig. 5(f)]. At $d = 40$ the two branches differ by 4.3 in energy. Snapping of the tether is the event in which the second core absorbs the trapped node ($q : 0 \to +1$). The twisted-moiré pair energies hardly change between $K/\mu = 0.5$ and 2: the ratio of dilational to shear strain in the twisted background is $3 \times 10^{-4}$, so the problem stays in the shear sector.

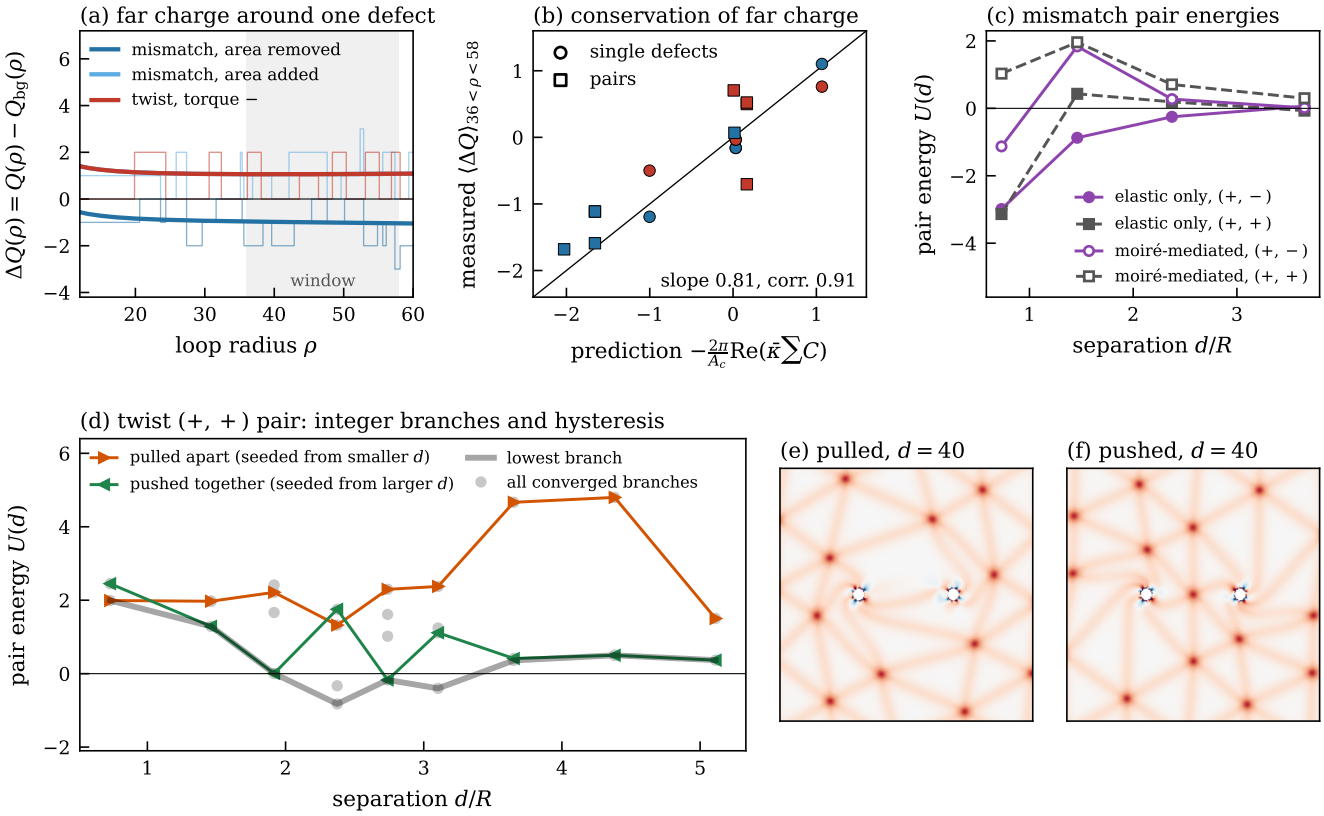



FIG. 5. Far charge and pair interactions. (a) Loop charge $\Delta Q(\rho)$ around single defects: integer data (thin steps) and linear prediction from $\sum C$ with box images (thick). The shaded window is used for averaging. (b) Measured versus predicted far charge for 8 single defects (circles) and 9 pair states (squares); blue: mismatch, red: twist. (c) Pair energy in the mismatch background: purely elastic part ($V_0 = 0$, filled) and moiré-mediated part (open) for closing–opening $(+,-)$ and closing–closing $(+,+)$ pairs. (d) Twisted $(+,+)$ pair: energies of all converged branches (gray), of the branch followed when pulling the cores apart (orange) and pushing them together (green), and the lowest branch. (e),(f) Local rotation $\omega = \frac{1}{2}\mathrm{curl}\, u$ of the pulled and pushed states at $d = 40$; walls appear as ridges. In (e) a doubled wall tethers the two cores.

## VI. THE DRESSED CHARGE

### A. Branches with equal atomic charge have different far charge

To separate the far field from the box boundary we embedded the small-box branches into boxes of half-width 148.5 and relaxed them again. Two branches with the same $\sum C$ have loop charges that differ out to $\rho \approx 140$ and disagree with the bare-charge prediction. Equation (11), however, evaluated with the measured $C_{\rm eff}(\rho)$, reproduces the loop charge within 0.1–0.2 for $\rho \geq 50$ in all four states. For $\rho \gtrsim 50$, $C_{\rm eff}(\rho)$ is fitted by a point source screened by the clamped edge, $C_0(1-\rho^2/L^2)$ with $L \approx 152$–$166$ [Fig. 6(a,b)]. The reorganization of the wall network is thus confined within $\sim 50$ of the pair, and farther out it appears only as a monopole $C_0 \neq \sum C$.

The difference $C_{\rm reg} = C_0 - C_{\rm bare}$ is an induced charge. Here $C_{\rm bare}$ is the dressed charge of a purely elastic run with identical boundary conditions and the same fitting form. It differs from the nominal 8 (9.71 for the twisted pair, 6.69 for the mismatch pair) because the rim penalty fixes the mean rim displacement rather than the far flux, and because of the clamped-box images; using the elastic run as reference removes these effects from $C_{\rm reg}$. In the twisted moiré a torque charge appears even though the bare torque is zero, and its sign differs between branches. In the mismatch moiré the area charge differs by 3.0 between branches (Table II). The far node charges of the two branches differ by 2.7 (twist) and 0.7 (mismatch). *The far charge is therefore not fixed by $\sum C$; the integer branch enters through $C_{\rm reg}$.*

### B. Force-dipole formula

The registry force $f = -\partial V/\partial u$ is a body force acting within the layer. When a defect reorganizes the wall network, the body force changes by $\Delta f$ relative to the background. If $\Delta f$ is localized and carries no net force, then in two-dimensional Lamé elasticity an isotropic force dipole $P\delta_{ij}$ produces $u_r = P/[2\pi(\lambda + 2\mu)r]$ and a point

torque $M$ produces $u_\theta = M/(4\pi\mu r)$ [27]. The traceless symmetric part produces $e^{\pm 2i\phi}$ fields that carry neither flux nor circulation. Comparing with $u_r = -\mathrm{Re}\, C/r$ and $u_\theta = -\mathrm{Im}\, C/r$ gives

$$C_{\mathrm{reg}} = -\frac{\mathrm{Tr}\, P}{4\pi(\lambda+2\mu)} - i\,\frac{M}{4\pi\mu}, \tag{12}$$

with $P_{ij} = \int x_i\, \Delta f_j\, \mathrm{d}^2x$ and $M = P_{xy} - P_{yx}$. The moiré lattice itself carries a periodic force distribution that cancels cell by cell. The moments must therefore be evaluated with a window that decays smoothly over more than one moiré period; a sharp cutoff produces large oscillations. With windows of radii 40–100 and 50–120, Eq. (12) reproduces the measured twist torque within 0.3 and the area part within 0.6 [Fig. 6(c)].

### C. Stiffness dependence of the induced charge

Equation (12) weights the area part by $1/(\lambda+2\mu)$. If the registry forces were fixed, the area part of $C_{\mathrm{reg}}$ would halve when $\lambda+2\mu$ doubles. We relaxed the same twisted branches at $K/\mu = 2$ ($\lambda = 1$, box half-width 109.8, with its own elastic reference). Equation (12) still holds at each stiffness, but the forces are not fixed. $\mathrm{Tr}\, P$ grows from 68 to 127 (pulled) and from 40 to 50 (pushed), and $\mathrm{Re}\, C_{\mathrm{reg}}$ changes by factors 1.07 and 0.87 instead of 0.5. The torque moments $M$ are almost unchanged ($-92/-79$ and $+42/+49$), as expected for a reorganization confined to the shear sector [Fig. 6(d)]. The evidence comes from one separation and two branches; for the pushed branch the growth of $\mathrm{Tr}\, P$ alone would predict a factor 0.63.

The registry reorganization therefore acts more like a source of prescribed *displacement*—the partial Burgers vectors carried by the walls—not of prescribed force. $C_{\mathrm{reg}}$ is a geometric quantity selected by the integer branch, and the force dipole is the elastic reaction needed to produce it. In the mismatch case at $K/\mu = 2$ the background develops a spontaneous chirality, which makes individual $C_{\mathrm{eff}}$ unreliable. The branch difference is still well defined: $\mathrm{Re}\,\Delta C_{\mathrm{reg}} = 2.6$ measured versus 2.55 from Eq. (12), close to the value 3.0 at $K/\mu = 0.5$.

### D. Integer bookkeeping and moiré Burgers circuits

Can $C_{\mathrm{reg}}$ be predicted from integer data? By the Gauss law, the $\kappa$-component of $C_{\mathrm{eff}}$ is a node count. Consider a smooth radial window $w(r) = \mathrm{clip}[(62-r)/32, 0, 1]$ and the window-weighted counts $N_{-,w}$ and $\Delta N_{+,w}$. Then

$$F \equiv -\frac{2\pi}{A_c}\mathrm{Re}(\bar\kappa C_{\mathrm{eff}}) = \sum_k Q_{\mathrm{hole},k} - N_{-,w} - \Delta N_{+,w}, \tag{13}$$

Here $F$ is the far node charge $\Delta Q$ of Eq. (11); Fig. 4(b) shows $-\Delta Q$ so that the far share appears positive. Equation (13) holds within 0.07 for all 50 pair states with $d \le 40$ [Fig. 7(a)]; the residual measures the linearization and the smoothing of the window. The bound integers $q_k$ and the core valences $z_k$ control the sign of $F$ in twisted pairs. With $\sum q = 1$ and one core of valence $z \le 6$, $F < 0$; with $\sum q = 1$ and both cores $z \ge 8$, or with $\sum q \ge 2$, $F > 0$ [Fig. 7(b)]. The magnitude is not fixed by $(q, z)$, because the network deficit $\Delta N_{+,w}$ varies by up to one node between states with equal $(q, z)$.

The missing integer is visible in moiré Burgers circuits. Since $u$ is continuous, the moiré Burgers vector $B = A^{-1}\oint \mathrm{d}u$ vanishes for every loop, so there are no true moiré dislocations. However, labeling every node by the AA lattice point it reaches and comparing labels with the rounded geometric steps along a Delaunay triangulation reveals geometric dislocations, i.e., 5–7-like defects of the node lattice. They occur as $\pm B$ dipoles, and only in the pulled (tethered) branches [Fig. 7(c)]. Their dipole moment $B \times s$ is an inserted moiré area. In Sec. VII we ask whether moiré elasticity converts it into far charge with a universal factor.

## VII. MOIRÉ ELASTICITY AND THE RING-TRACTION IDENTITY

### A. Elastic constants of the moiré crystal

We define the Eulerian moiré displacement $U$ by $u = \kappa(z - U)$ and impose homogeneous strains $U = U_a z + U_b \bar z$ on oblique periodic cells spanned by the strained moiré vectors ($2\times2$ moiré cells). We then fit the relaxed energy density as

$$e = e_0 + \sigma_d\varepsilon_d + 2K_M\varepsilon_d^2 + \tfrac{1}{2}\chi_M\omega^2 + 2\mu_M|\gamma|^2 + \ldots, \tag{14}$$

with $\varepsilon_d = \mathrm{Re}\, U_a$, $\omega = \mathrm{Im}\, U_a$ and $\gamma = U_b$ (35 strain states, residual $< 10^{-8}\theta^2$). All terms forbidden by hexagonal symmetry—linear shear, anisotropy, and the $\varepsilon_d\omega$ cross term—vanish within numerical precision. The results (Table III, Fig. 8) differ qualitatively from the unrelaxed values. A moiré dilation is an atomic rotation in the twisted case and an atomic dilation in the mismatch case, which fixes the bare entries.

Two signs stand out. The twisted moiré has a *negative bulk modulus*: a uniform moiré dilation (a local untwisting) lowers the energy. It satisfies strong ellipticity ($\mu_M > 0$, $K_M + \mu_M = 1.000\,\theta^2 > 0$) and is therefore stable under clamped boundaries, but unstable to homogeneous dilation when the boundary is free. That $K_M + \mu_M$ equals the unrelaxed value is a numerical observation we cannot yet explain. The mismatch moiré has a *negative rotational stiffness*: it tends to rotate spontaneously, consistent with the small global rotation observed in mismatch backgrounds. In clamped boxes the bulk moiré expands uniformly, $\varepsilon_0 \approx \Delta^*/L$ with $\Delta^* \approx 2.7$–$3.0$ for the twist [Fig. 8(c)]. The excess twist is stored in an edge layer near the clamp. Treating the bulk as uniform and the edge layer as a line energy $g(\Delta)$ of the mismatch $\Delta = (\varepsilon_b - \eta)L$, the equilibrium condition

TABLE II. Dressed charge in large boxes ($K/\mu = 0.5$). $q$: integers bound by the two cores; $C_0$: fitted monopole; $C_{\rm reg} = C_0 - C_{\rm bare}$ measured and from Eq. (12); $F$: far node charge $-(2\pi/A_c)\mathrm{Re}(\bar\kappa C_0)$. In the mismatch case only the area part enters $F$.

| state | $q$ | $C_0$ | $C_{\rm reg}$ | $C_{\rm reg}$ [Eq. (12)] | $F$ |
|---|---|---|---|---|---|
| twist, pulled | (1,0) | $6.73 + 7.55i$ | $-2.98 + 7.55i$ | $-3.6 + 7.3i$ | $-1.83$ |
| twist, pushed | (1,1) | $8.11 - 3.70i$ | $-1.60 - 3.70i$ | $-2.1 - 3.4i$ | $+0.90$ |
| mism., pulled | (0,−1) | 7.11 | $+0.42$ | $0.0 \pm 1$ | $-1.72$ |
| mism., pushed | (0,0) | 4.10 | $-2.59$ | $\approx -3.0$ | $-0.99$ |

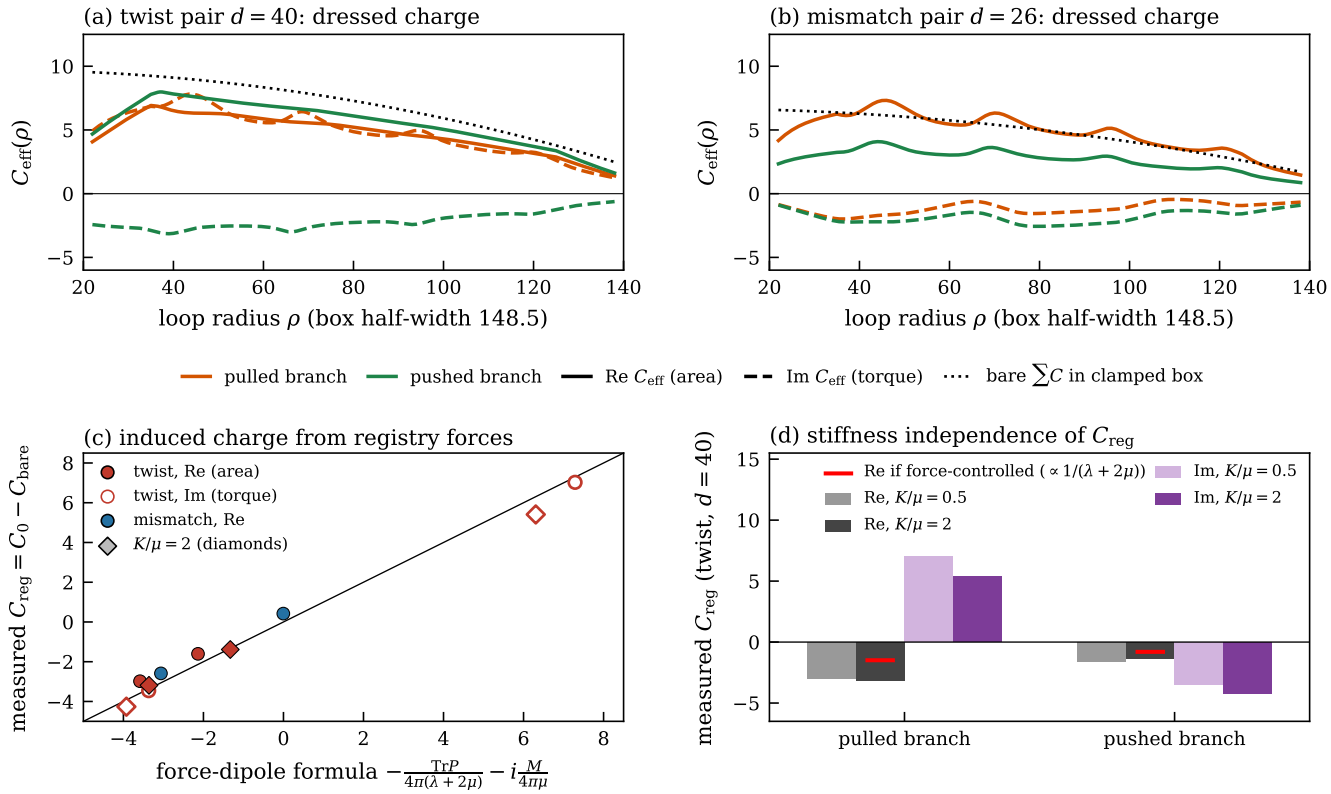



FIG. 6. Dressed charge in large boxes (half-width 148.5). (a),(b) Dressed charge $C_{\rm eff}(\rho)$ of the pulled and pushed branches of a twisted pair ($d = 40$) and a mismatch pair ($d = 26$), smoothed over one moiré period. Solid lines: area part; dashed lines: torque part. The dotted line is the bare charge in the clamped box. (c) Induced charge $C_{\rm reg}$ measured versus the force-dipole formula, Eq. (12), for $K/\mu = 0.5$ (circles) and $K/\mu = 2$ (diamonds); filled: area part, open: torque part. (d) $C_{\rm reg}$ of the twisted pair at two stiffnesses. The red bars mark the area part expected if the registry forces were fixed ($\propto 1/(\lambda + 2\mu)$).

TABLE III. Moiré elastic constants in units of $\theta^2$ ($\lambda = -0.5$, $\mu = 1$). The last row gives the values without registry relaxation (twist/mismatch).

| | $K_M$ | $\mu_M$ | $\chi_M$ | $\sigma_d$ | $K_M/(K_M + \mu_M)$ |
|---|---|---|---|---|---|
| twist | $-0.149$ | 1.149 | 6.41 | $-3.34$ | $-0.149$ |
| mismatch | 0.273 | 0.479 | $-1.29$ | $-5.92$ | 0.363 |
| bare | 0/0.5 | 1/1 | 2/0 | – | 0/0.33 |

$g'(\Delta) = -\sigma_d/2 - 2K_M\varepsilon_b$ is dominated by $\sigma_d$. This explains the $1/L$ law. The edge is nevertheless stiff: when a dilation $\eta$ is imposed on the clamp, the bulk follows it almost one to one [Fig. 8(d)].

### B. A local moiré source

If these constants governed local sources, a disk of inserted moiré area $\delta A^* = 2\pi r_0^2 \varepsilon^*$ would send the fraction $S = K_M/(K_M + \mu_M)$ of it to infinity, in analogy with an Eshelby inclusion [26]. The far factor would then be $f = 1 - S$: 1.15 for the twist and 0.64 for the mismatch. To test this we construct the moiré analog of an eigenstrain inclusion. Inside a disk $D$ of radius $r_0$ cen-

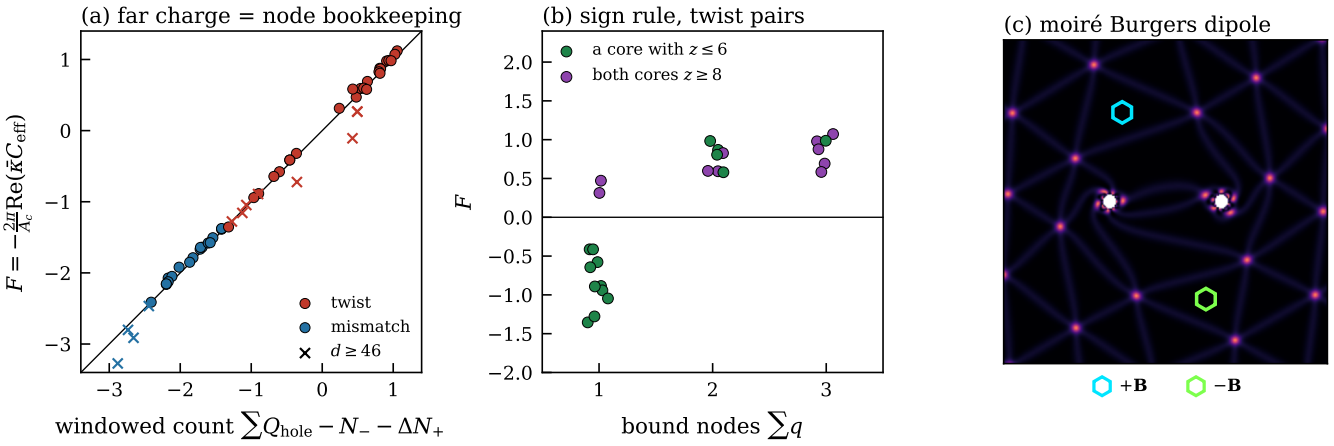



FIG. 7. Integer bookkeeping. (a) Far charge $F$ versus the window-weighted integer count of Eq. (13) for 53 pair states (crosses: $d \geq 46$, cores close to the window edge). (b) Sign rule for twisted pairs ($16 \leq d \leq 46$): $F$ versus the total bound charge $\sum q$, colored by the smallest core valence. (c) Geometric moiré Burgers circuits in a pulled twisted pair ($d = 40$): hexagons mark Delaunay triangles of nodes with a non-closing circuit; their $\pm B$ form a dipole.

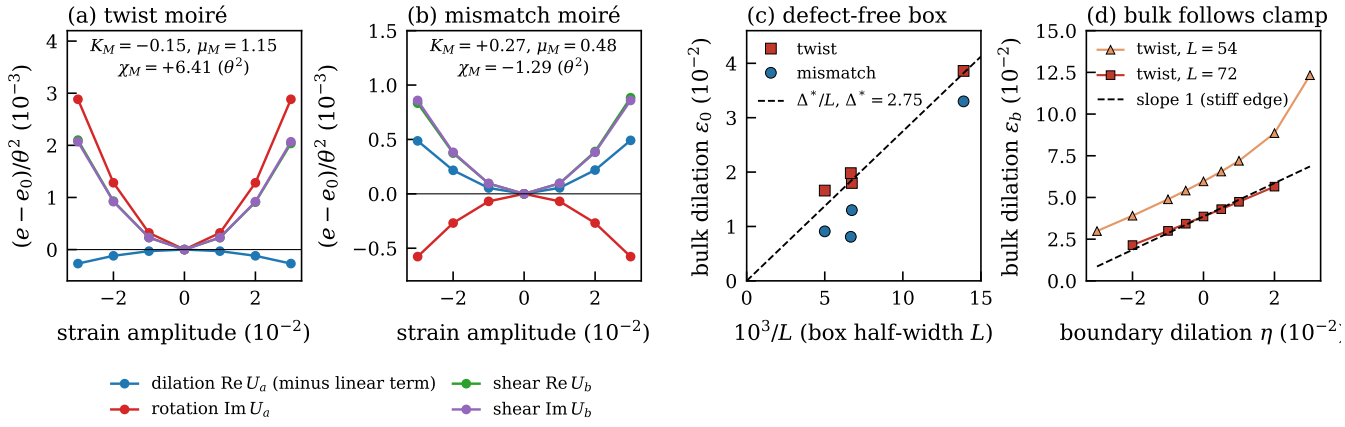



FIG. 8. Elasticity of the moiré crystal. (a),(b) Relaxed energy density versus homogeneous moiré strain for the twisted and mismatch moiré (periodic cells). The linear pre-stress term $\sigma_d \varepsilon_d$ has been subtracted from the dilation curve; the fitted constants are given in units of $\theta^2$. (c) Uniform bulk dilation $\varepsilon_0$ of defect-free clamped boxes versus inverse box half-width. (d) Bulk dilation of twisted boxes versus a dilation $\eta$ imposed on the clamped boundary; the dashed line has slope one.

tered at $z_0$, the energy is evaluated at $u + s$ instead of $u$, with $s = \varepsilon^* \kappa (z - z_0) \mathbb{1}_D$. The registry potential becomes $V(u + s)$ and $\varepsilon^* \kappa$ is added to the elastic gradient. The natural state inside is a moiré dilated by $\varepsilon^*$, and the jump of $s$ at $\partial D$ is a ring of fractional moiré dislocations.

The far response, measured from exact node positions, does not follow $S$. In twisted and mismatch moirés with $r_0 = 45$–$95$ (1.5–3.2 moiré periods) in boxes of half-width 150 and 200, $S = A/(\varepsilon^* r_0^2)$ ranges from $-8$ to $+5.5$. Here $A$ is the coefficient of the far field $A(1/r - r/L_{\rm eff}^2)$, whose effective radius $L_{\rm eff}$ always matches the box. The value of $S$ depends on $r_0$, on whether the ring passes through a node shell, and on the sign of $\varepsilon^*$ [Fig. 9(d)].

### C. Ring-traction identity

The drive of such a source obeys an exact identity. Let $u_0$ be the equilibrium at $\varepsilon^* = 0$ and $w = \kappa(z - z_0)$. By the envelope theorem,

$$E_1 \equiv \frac{\partial E}{\partial \varepsilon^*}\Big|_0 = \int_D \sigma_{\rm at} : \nabla w \, {\rm d}A + \int_D \nabla V(u_0) \cdot w \, {\rm d}A,$$

where $\sigma_{\rm at}$ is the atomic Cauchy stress of $u_0$. The equilibrium condition $\nabla V(u_0) = {\rm div}\, \sigma_{\rm at}$ and one integration by parts cancel the bulk terms exactly:

$$E_1 = \oint_{\partial D} (\sigma_{\rm at}\, n) \cdot \kappa \, (z - z_0) \, {\rm d}l. \qquad (15)$$

The local source therefore couples only to the microscopic traction on its boundary [Fig. 9(a)]. For the twist, $\kappa(z - z_0)$ is tangential and $E_1 = \theta r_0 \oint t_t \, \mathrm{d}l$ is the microscopic torque on the ring. Averaged over the position $z_0$ within a moiré cell, the mean stress is symmetric and $\langle E_1 \rangle = 0$. For the mismatch, $\kappa(z - z_0)$ is radial and $E_1 = \theta r_0 \oint t_n \, \mathrm{d}l$. The mean atomic stress produced by the background strain $\theta$ gives $\langle E_1 \rangle = 4(\lambda + \mu)\theta^2 \pi r_0^2$, which equals $2\theta^2 \pi r_0^2$ here. A continuum moiré description would instead predict $E_1 = -\pi r_0^2 \sigma_d$ (+57 and +101 at $r_0 = 70$ for twist and mismatch). This coupling is absent. The mismatch mean exists because the total-$u$ reference stresses the mismatch background; in a stress-free reference it would vanish as well.

Table IV and Fig. 9(b,c) verify Eq. (15). Evaluated on the baseline field alone, it reproduces the finite-difference drive of the inclusion runs within 1–2% in the linear regime ($|\varepsilon^*| \leq 0.01$). Averages over 16 ring centers agree with the predicted means within one standard error. The fluctuation between placements grows faster than $r_0$, with a standard deviation of about 0.3 of the mismatch mean, and the centered, sixfold-symmetric ring is always an extreme. The negative-$\varepsilon^*$ jumps seen in the mismatch runs are rearrangements of nodes next to the ring, where the drive is concentrated.

### D. Consequence

Equation (15) concerns the drive, not the response: the far coefficient $S$ also depends on how the surrounding moiré responds, and for $r_0 \leq 3.2 L_M$ no homogenized description is expected. The identity nevertheless explains why $S$ scatters with placement and why it does not follow $K_M/(K_M + \mu_M)$. The placement-averaged drive is universal, but it is set by the atomic background stress rather than by the moiré pre-stress $\sigma_d$. The same reasoning applies to every moiré-scale defect embedded in a continuous registry, including the geometric $\pm B$ dipoles of Fig. 7(c): they couple through the forces their cores exert on the atomic layer. For the far node charge we therefore rely on the exact accounting of Eq. (13) together with the force-dipole dressing of Eq. (12); this is an accounting, not yet a predictive theory of $C_{\mathrm{reg}}$. Whether a placement-averaged $S$ converges for $r_0 \gg L_M$ remains open. The moiré constants of Table III describe homogeneous strains, such as the boundary-driven and $1/L$ bulk dilations of Fig. 8.

TABLE IV. Ring-traction identity, Eq. (15). $E_1$ from the ring integral on the baseline field, from finite differences of the inclusion runs, and the mean ± standard deviation over 16 ring centers, compared with the predicted mean.

| | $r_0$ (box) | identity | runs | 16 centres | pred. |
|---|---|---|---|---|---|
| twist | 45 (150) | −8.82 | −8.80 | $-0.4 \pm 2.9$ | 0 |
| twist | 70 (150) | −16.94 | −16.84 | $-1.7 \pm 9.0$ | 0 |
| mism. | 45 (150) | 0.88 | 0.90 | $14.1 \pm 4.2$ | 14.1 |
| mism. | 60 (150) | 33.10 | 32.6 | $26.5 \pm 7.4$ | 25.1 |
| mism. | 70 (150) | 7.65 | 7.76 | $32.2 \pm 11.2$ | 34.2 |
| mism. | 95 (200) | 21.99 | – | $60.8 \pm 21.2$ | 63.0 |

## VIII. LENSES LARGER THAN THE MOIRÉ PERIOD

The simulations above have $R \approx 11 < L_M \approx 30$ and $T = 0.97$, at the boundary of regimes I and II defined below; each defect removes about one node. What happens when the lens is larger than the moiré lattice constant? Three lengths enter. The atomic core $r_c \simeq \sqrt{|C|}$ is where the defect strain $|\nabla \delta u| = |C|/r^2$ reaches order one. The Einstein radius $R = \sqrt{|C|/\theta}$ is where the defect strain equals the background registry gradient $\theta$. The moiré period is $L_M = a/\theta$. Their ratios are

$$\frac{r_c}{R} = \sqrt{\theta}, \qquad \frac{R}{L_M} = \sqrt{\frac{T A_c}{2\pi}}, \qquad T = \theta n_{\mathrm{miss}}. \tag{16}$$

Since the atomic strain at the Einstein ring equals $\theta \ll 1$, linear elasticity of the layer, and hence the lens map, holds everywhere outside $r_c$ regardless of the lens size. Large lenses are in fact the cleanest. Equation (16) defines four regimes [Fig. 10(d)]: (I) $T < 1$, where the far distortion is a fraction of a node; (II) $1 < T < 2\pi/A_c \approx 7$, the few-node lens with $R < L_M$; (III) $R > L_M$ but $r_c < L_M$, the resolved lens; and (IV) $n_{\mathrm{miss}} > 2\pi/(A_c \theta^2)$, where the core itself exceeds the moiré period.

*Arcs and the moiré Einstein ring.* In regime III the critical circle $|z| = R$, where $J = 0$, is resolved by many nodes. It maps onto the caustic $|\beta| = b_s$ of the source plane. Node images close to it are stretched tangentially into arcs, exactly like lensed galaxies [Fig. 10(a)]. When $\varphi_m = 0$ the caustic shrinks to a point. On $|z| = R$ one has $\beta = z - R^2/\bar{z} = 0$, so the entire circle carries the single registry value of the undeformed pattern at the defect position. The result is a closed ring of uniform stacking, a *moiré Einstein ring* [Fig. 10(b)]. Its stacking is AA if the defect sits on an AA site and AB if it sits on an AB site, so the ring images the registry at the defect position. In a mismatch bilayer, removing area from the larger-lattice layer (or adding area to the smaller-lattice layer) gives $\varphi_m = 0$ and such a ring once $T \gtrsim 7$; the opposite combinations give $\varphi_m = 180°$ and swallow the maximal disk $b_s = 2R$. In a twisted bilayer a ring requires $\varphi_m = 0$, i.e., a torque charge; an area defect produces arcs and swallows the nodes inside $b_s = \sqrt{2}R$.

*Inverted moiré inside the ring.* Inside $R$ every source has an inverted image, so the region $r_c < |z| < R$ hosts a mirror-image moiré. Its node density $n_M R^4/|z|^4$ grows toward the core, and its total node number $\pi n_M R^4 (r_c^{-2} - R^{-2})$ is of order $\pi |C|/A_c = n_{\mathrm{miss}}/2$, independent of $\theta$ (the prefactor depends on the core cutoff). This inner pattern is the moiré generated by the defect's own strain

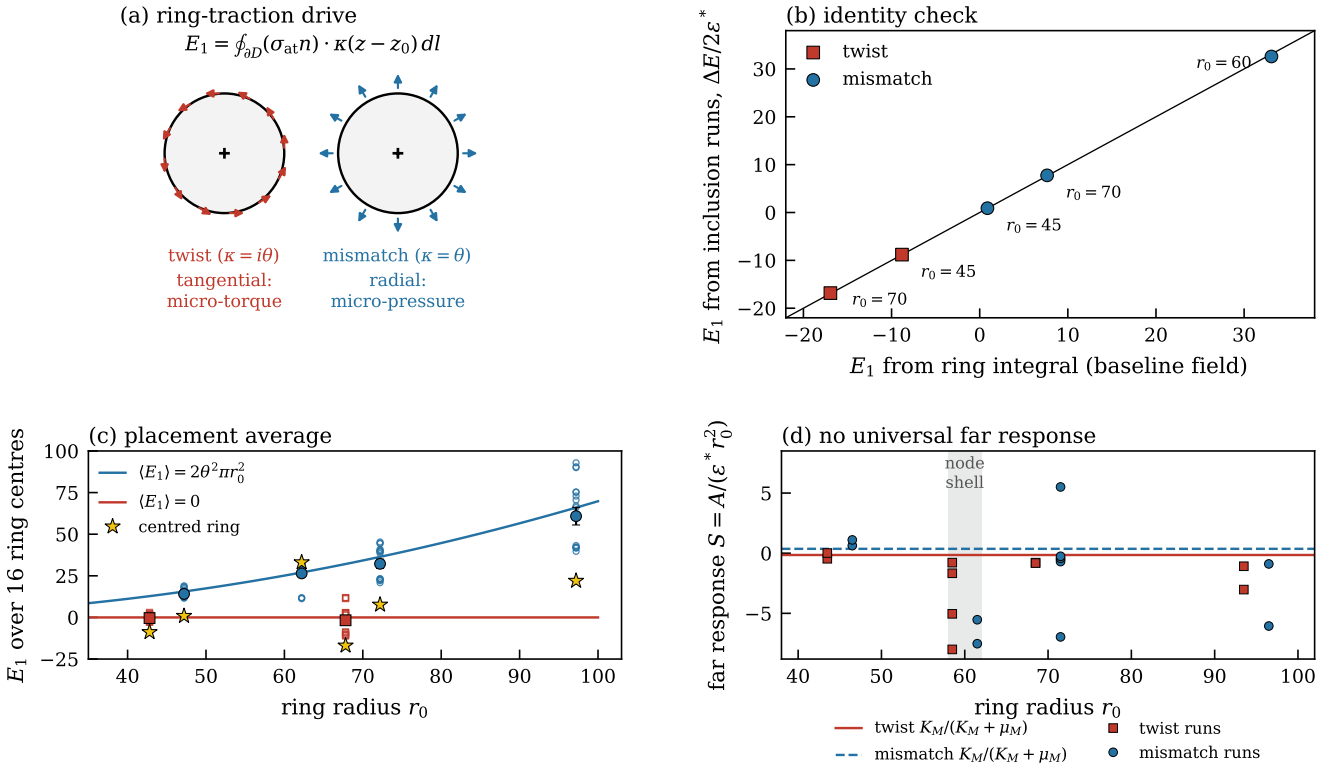



FIG. 9. Ring-traction identity. (a) The drive of a local moiré source is the atomic traction integrated against $\kappa(z-z_0)$ on its boundary: tangential (micro-torque) for the twist, radial (micro-pressure) for the mismatch. (b) $E_1$ from Eq. (15) on the baseline field versus the finite-difference drive of the inclusion runs. (c) $E_1$ for 16 ring centers per radius (small symbols), their mean ± standard error, and the centered ring (stars), compared with the predicted means 0 (twist) and $2\theta^2\pi r_0^2$ (mismatch). (d) Far response $S = A/(\varepsilon^* r_0^2)$ of all inclusion runs (single placements). No common value emerges, in particular not the Eshelby value $K_M/(K_M+\mu_M)$ (horizontal lines). The gray band marks a ring lying on a node shell.

field. Outside the caustic disk, the swallowed number $\pi n_M b_s^2 = T(1-\cos\varphi_m)$ becomes large. For the linear lens map the lattice-point counting error scales with the perimeter, so the relative rounding error decreases as $L_M/b_s$. Relaxation, however, can shift the bound integer by more than this (see below).

*Cores larger than the moiré period.* In regime IV the defect contains its own moiré. A grain of radius $r_g$ rotated by $\phi$ has $C = -i\phi r_g^2$ outside and a uniform local twist $\theta+\phi$ inside, so $R = r_g\sqrt{|\phi|/\theta}$. For $|\phi| < \theta$ the Einstein circle lies inside the grain and the grain is a smooth twist-angle domain. For $|\phi| > \theta$ the critical circle lies outside the grain. In particular, a grain rotated by $\phi = -2\theta$ has local twist $-\theta$ inside and $\varphi_m = 0$, and is surrounded by an Einstein ring at $R = \sqrt{2}\,r_g$ [Fig. 10(c)]. This assumes a grain whose rotation is imposed and elastically transmitted to the surroundings. A grain bounded by a relaxed low-angle grain boundary has no long-range field, and the smooth twist-angle variations of real samples [14] are not discrete grains. Where the exterior field is $-C/\bar{z}$, however, the node accounting of Eq. (10) applies directly.

*Relaxation.* Figure 10(e,f) shows relaxed continuum states for the defect of panels (a,b) ($|C| = 120$, $r_h = 20$, box half-width 150). Relaxation keeps the lens structure but converts it into domain walls, and the two backgrounds relax differently.

In the twisted bilayer, the lens map places ordinary images down to the critical circle $r = R = 60$, where they are compressed into arcs. After relaxation no ordinary node survives inside $r \approx 85 \approx 1.4R$: the arcs dissolve, and a fan of spiral walls connects the core to the outer network. The core binds $q = Q_{\rm hole} - N_- = 37$ nodes, compared with the lens value $T = 29$. Every circle in the node-free annulus crosses exactly $z = 42$ walls. If the 37 enclosed AA points carry unit winding, Eq. (6) requires $z \geq 2L_{\rm min}(37) = 42$, since a compact hexagon of 37 lattice points meets 21 lattice lines. The relaxed large lens attains exactly this minimum. The extra bound nodes appear as dressed far charge, $\Delta Q(\rho) \approx 6$–$12$ at $\rho = 100$–$130$ instead of the unscreened linear 3–5. Since the box half-width is only $2.5R$, this comparison is semi-quantitative, and each case is a single relaxed state.

In the mismatch bilayer the AA Einstein ring costs registry energy and does not survive. The ring and the

TABLE V. Defect hierarchy under $u = \kappa U$.

| level | atomic | moiré | in own units |
|---|---|---|---|
| disclination | $s \in 60^\circ\mathbb{Z}$ | same $s$ | integer |
| dislocation | $b \in \Lambda$ | $B = A^{-1}b \in \Lambda_M$ | integer |
| area/torque | $C$ (complex) | $m = C/\kappa$ | $\times\theta$ (fractional) |

inverted inner moiré reconstruct into nested closed domain walls bounded by the Einstein circle, i.e., a target pattern of commensurate annuli. Here the core binds $q = 1$ (lens value 0), and the ordinary-node deficit follows the lens prediction ($\Delta N_+ \approx 17$–$23$ against 18–26). Thus $\varphi_m$ selects the relaxed texture: a lens that rotates its images ($\varphi_m = \pm 90^\circ$) relaxes into a spiral wall fan, and a lens that only dilates them ($\varphi_m = 0$) relaxes into concentric walls. When the defect sits on an AB site, the Einstein ring is already a low-energy stacking and should survive relaxation as a commensurate annulus; we have not computed this case.

*Transmitted charge.* A vacancy cluster lenses only through the part of its area that the surrounding layer actually closes. A large hole with free edges transmits little $C$, whereas inserted islands and rotated grains always transmit their full charge.

## IX. DISCUSSION

*Hierarchy of charges.* The affine map $u = \kappa U$ relates atomic and moiré defects level by level (Table V). Disclinations and dislocations are homotopy invariants ($\pi_1(T^2) = \mathbb{Z}^2$ and the point group). Because $A$ maps lattice to lattice, an integer atomic invariant becomes an integer moiré invariant. The atomic area charge is quantized by the number of atoms but is not a homotopy invariant ($\pi_2(T^2) = 0$), whereas the moiré node number is a degree. The continuous parameter $\theta$ sits in between, so the far share $\theta n_{\mathrm{miss}} \cos\varphi_m$ is fractional, and the integer local share of the nonlinear core makes up the difference. Writing the node density as the Jacobian $(1/4\pi^2)\epsilon^{ab}\partial_a\phi_1\partial_b\phi_2$ of the two moiré phases $\phi_i = \boldsymbol{G}_i \cdot u$ makes the analogy with Goldstone–Wilczek induced charge explicit [28].

*Twist versus mismatch.* The two backgrounds differ only in $\alpha$, but the consequences are distinct. (i) Area defects are swallowed with $b_s = \sqrt{2}R$ in a twisted moiré and not at all in a mismatch moiré. Frustrated integer assignment in the twisted case produces multiple branches and hysteresis. (ii) Relaxation of the twisted moiré is rotational and its walls are shear walls, so the bulk modulus drops out of the defect problem. Mismatch relaxation is dilational and sensitive to $K$. (iii) The mismatch background is mirror symmetric, so node rotation is an Ising-like variable; this is consistent with $\chi_M < 0$ and with the spontaneous chirality at large $K$. In the twisted background the mirror is absent and node rotation is pinned.

*Heterostrain.* A uniform heterostrain adds $\eta\bar{z}$ to the background, $u_b = c_{\mathrm{AA}} + \kappa z + \eta\bar{z}$. The lens map becomes $\beta = z + (\eta/\kappa)\bar{z} - m/\bar{z}$, the point-mass lens with external shear (the Chang–Refsdal lens [35]). Its caustic is an astroid instead of a circle, and regions with four images appear. The relevant parameter is $|\eta/\kappa| \sim \epsilon_h/\theta$ for a heterostrain $\epsilon_h$, so even a heterostrain of 0.1% gives $|\eta/\kappa| \approx 0.05$ near the magic angle, and the lens is distorted more strongly at smaller $\theta$. The partition then acquires an angular modulation; we leave this to future work.

*Experimental consequences.* The lens scales follow from Eq. (16) without model input. A defect that removes $T$ nodes has $R/L_M = \sqrt{TA_c/2\pi} \approx 0.37\sqrt{T}$, independent of $\theta$; for $T = 1$ it must miss $1/\theta$ unit cells (about 52 at $\theta = 1.1^\circ$). A single removed node is therefore resolved as a lens only when $0.37L_M$ exceeds the domain-wall width, which in our model ($w \approx 0.13L_M$) is satisfied. In real bilayers the wall width is fixed in nanometres [7], so the condition favors small twist angles. Independently of resolving the swallowed region, the lens displaces the surrounding nodes by $|m|/r = R^2/r$; at $r = L_M$ this is $0.14\,TL_M$, about 1.8 nm per removed node at $\theta = 1.1^\circ$, which is within reach of drift-corrected STM or interferometric moiré metrology [14, 15, 17, 18]. The resolved regime $R > L_M$ requires $n_{\mathrm{miss}} > 2\pi/(A_c\theta) \approx 7.3/\theta$ unit cells, about 380 at 1.1°. A real vacancy cluster lenses only with the charge its surroundings actually close; multivacancy reconstruction and buckling reduce $C$ below $n_{\mathrm{miss}}A_c/2\pi$ [20]. Our mismatch case uses a homobilayer registry with degenerate AB and BA stackings. A heterostructure such as graphene on hBN has inequivalent stackings, so our predictions for it are qualitative: an area defect lenses with $\varphi_m = 0$ or 180° depending on the host layer, and whether the Einstein ring of a $\varphi_m = 0$ lens survives relaxation depends on the energy of the stacking it carries. Finally, two defects with the same atomic content can differ in far node charge by up to $\sim 3$ nodes in our simulations (one pair geometry), so the moiré around a defect carries information about its core state.

*Model assumptions.* The topological statements (Secs. III and IVA) hold for any continuous registry map with vanishing Burgers class. The partition theorem relies on the linear region outside the core, i.e., on $R$ being larger than the wall width. It also requires that the registry forces produce no net stress on average over the moiré lattice. The single relative-displacement field carries the stiffness of the bilayer combination; how a defect's field is shared between the layers rescales $C$ but not the lens structure. The elastic parameters, in particular the auxetic $K/\mu = 0.5$, are model choices; the area part of the induced charge of the twisted pair changes by less than 15% between $K/\mu = 0.5$ and 2. The single-field description neglects layer-resolved displacement and out-of-plane relaxation; these change $V$ and the elastic constants but not the topology or the lens structure. The numerical branch search is not exhaustive, and the nudged-elastic-band barriers [30] we computed for branch

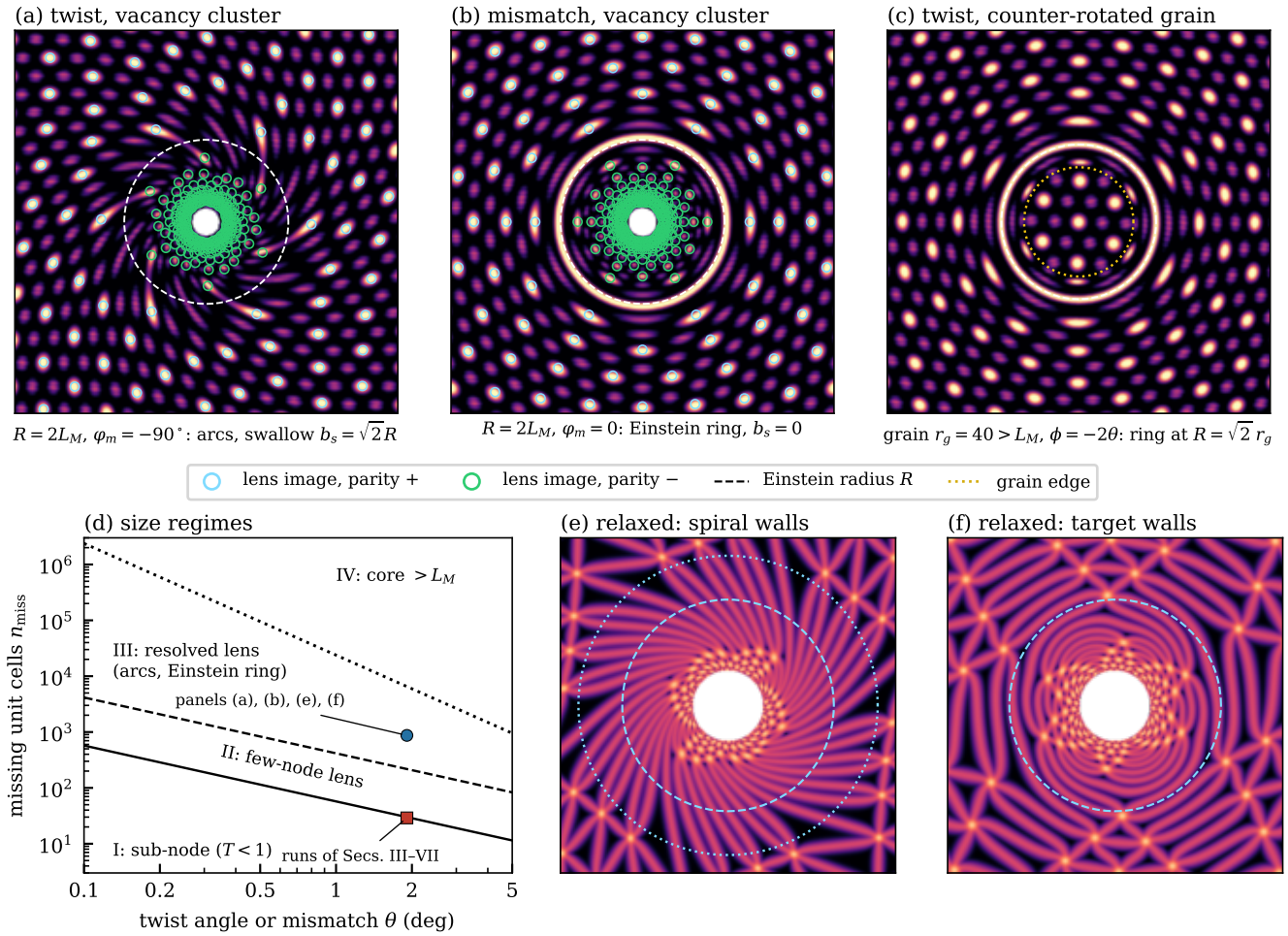



FIG. 10. Lenses larger than the moiré period ($\theta = 1/30$, $L_M = 30$). (a)–(c) Moiré computed from atomic positions as in Fig. 1. (a) Vacancy cluster with $|C| = 120$ ($R = 2L_M$) in a twisted bilayer: images near the critical circle are stretched into arcs, and the region inside $b_s = \sqrt{2}R$ is swallowed. (b) The same defect in a mismatch bilayer ($\varphi_m = 0$): the whole critical circle carries the registry of the defect position (here AA), forming a moiré Einstein ring; inside it the moiré is inverted (green). (c) A grain of radius $r_g = 40 > L_M$ (yellow dotted) rotated by $\phi = -2\theta$ in a twisted bilayer: the grain hosts its own moiré of twist $-\theta$, and an Einstein ring forms at $R = \sqrt{2}r_g$. (d) Size regimes of Eq. (16) in the plane of twist angle and missing unit cells. Square: defects of Figs. 2–9 ($T = 0.97$, at the I/II boundary); circle: panels (a), (b), (e), (f). (e),(f) Relaxed continuum calculation of the defect of (a) and (b) ($r_h = 20$, box half-width 150), shown as $\log_{10} V(u)$ so that domain walls appear as bright lines; dashed: Einstein radius $R$; dotted in (e): $\sqrt{2}R$. The twisted lens relaxes into a fan of 42 spiral walls around a core binding 37 nodes, which saturates the bound of Eq. (6); the mismatch lens relaxes into nested closed walls (target pattern) inside the Einstein circle.

switching are upper bounds.

## X. CONCLUSIONS

Moiré nodes are topological charges of the registry map, and atomic area and torque defects act on them as lenses with mass $m = C/\kappa$. On average over its position, a defect missing $n_{\rm miss}$ unit cells removes $\theta n_{\rm miss}$ ordinary nodes; a fraction $1 - \cos\varphi_m$ is swallowed and bound to the core as an integer, and a fraction $\cos\varphi_m$ is pushed to infinity. This explains why area defects are swallowed in twisted bilayers but, depending on the host layer, not in mismatch bilayers, and why closing and opening irises are symmetric in one case and not in the other. Beyond the atomic charge, the far node charge is dressed by the force dipole of the reorganized registry forces; this induced charge depends on the integer branch and responds to stiffness more like a displacement-controlled source. The moiré crystal has anomalous elastic constants—a negative bulk modulus for twist and a negative rotational stiffness for mismatch—yet the drive of a local source is the microscopic ring traction rather than the moiré pre-stress. When the Einstein radius exceeds the moiré period the lens is resolved into arcs, an inverted inner moiré and, for $\varphi_m = 0$, a ring of uniform stacking; re-

laxation turns these into spiral or concentric wall textures, enlarges the swallowed region, and dresses the far charge. Open problems include an integer-resolved partition theorem tested over a range of $|C|$, a rule that predicts $C_{\rm reg}$ from the core state, placement-averaged responses for sources much larger than $L_M$, heterostrained lenses, and the node currents pumped by moving defects.

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## DATA AVAILABILITY

Code and analysis data are available at https://github.com/cywon-commits/moire-iris-lens.

### Appendix A: Equivalence of elastic references

The clamped calculations assign elastic energy to the strain of the total $u$, i.e., $E_{\rm el}[u] = E_{\rm el}[u_b] + E_{\rm el}[\delta u] + \int \sigma_b : \varepsilon(\delta u)$. For affine $u_b$ the stress $\sigma_b$ is uniform, and the cross term becomes the boundary integral $\oint \sigma_b n \cdot \delta u$. It vanishes on the outer boundary ($\delta u = 0$). On a core rim $\sigma_b$ is isotropic (mismatch) or zero (twist), so the cross term couples only to the mean normal displacement, which is fixed by the constraint. The cross term is therefore a constant, and equilibria, energy differences and interactions of iris defects agree between the two references. This equivalence does not extend to the eigen-inclusion of Sec. VII, whose jump $s$ couples to the background stress through $\int_D \sigma_b : \nabla s$; this term is exactly the placement-averaged mismatch drive $\langle E_1 \rangle$, which vanishes in a stress-free reference.

### Appendix B: Removed node charge of a pair

For two defects, $J = 1 - |f'|^2$ with $f' = -m_1/(\bar{z} - \bar{z}_1)^2 - m_2/(\bar{z} - \bar{z}_2)^2$. Averaged over translations, the number of removed ordinary nodes is $n_M \int (1 - J\mathbb{1}_{J>0}) \mathrm{d}^2 z = n_M \int \min(1, |f'|^2) \mathrm{d}^2 z$. For one defect this gives $2\pi R^2 n_M = T$, consistent with Eq. (10). Numerically, $(+,+)$ pairs give 1.79–1.93, and $(+,-)$ pairs give $0.02, 0.93, 1.46, 1.90$ at $d = 0, 8, 16, 30$.

### Appendix C: Counting

*Nodes:* on each grid cell, we count the windings of the polygon formed by the registry values at its four corners around the AA lattice points. Shared edges cancel, so Eq. (5) holds exactly on the grid. $Q_{\rm hole}$*:* winding along a staircase loop just outside the core (radius $\approx 3$). $z$*:* changes of the AB/BA label along circles of radius 5, 7 and 9. *Node positions* for the inclusion analysis are obtained by solving $u(z) = c_{\rm AA} + n$ with Newton iterations on a cubic interpolant (residual $< 10^{-12}$).

### Appendix D: Numerical parameters

TABLE VI. Parameters (lattice constant $a = 1$).

| | |
|---|---|
| twist angle or mismatch $\theta$ | 1/30 |
| registry $V_0$, $V_3$ | 0.005, $\sqrt{3}V_0$ |
| Lamé $\mu$; $\lambda$ | 1; $-0.5$ or 1 |
| $K/\mu = (\lambda + \mu)/\mu$ | 0.5 or 2 |
| $|C|$; core radius $r_h$; rim penalty | 4; 2.4; 200 |
| Einstein radius $R = \sqrt{|C|/\theta}$ | $\approx 11$ |
| grid spacing $h$ | 0.45 |
| box half-width | 54–200 |
| relaxed moiré period | $\approx 31$ |


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